

Task Scheduler System Architecture

Patterns: Observer Pattern, Command Pattern, Strategy Pattern (Scheduling), Worker Pool | Domain: Concurrency & Scheduling / Task Engine

Core Requirements & Features	Core Domain Classes & Interfaces
• Task Execution Modes: One-time tasks, Recurring fixed-rate tasks, Cron expression tasks. • Priority Scheduling: Execute high-priority tasks ahead of low-priority tasks. • Preemption & Cancellation: Cancel scheduled tasks, handle execution failures & retries.	• CPUScheduler: Central engine managing task queue and worker threads. • ScheduledTask: Runnable/Callable wrapper with execution time, priority, & frequency. • SchedulingAlgorithm: Strategy interface (FCFS, Priority, SJF, Round Robin). • Dispatcher: Worker thread pool pulling ready tasks from priority queue.

Critical Execution Workflow & Method Trace

1. Client calls `schedule(task, delay, period, TimeUnit)`.
 2. Calculates initial `executionTime = currentTime + delay`.
 3. Wraps payload into `ScheduledTask` object with priority and interval.
 4. Inserts `ScheduledTask` into `DelayQueue` / `PriorityBlockingQueue` sorted by `executionTime`.
 5. Dispatcher worker thread calls `queue.take()` (blocks until head task's delay expires).
 6. Worker thread executes `task.run()` inside Worker Thread Pool.
 7. If task is Recurring: Resets `executionTime = currentTime + period` and re-inserts `ScheduledTask` back into `PriorityBlockingQueue`.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• DelayQueue Safety: Thread-safe <code>PriorityBlockingQueue</code> with <code>ReentrantLock</code> & <code>Condition</code> variables. • Worker Thread Isolation: <code>UncaughtExceptionHandler</code> catches worker thread exceptions to prevent pool crashes. • Task Cancellation: Volatile boolean <code>isCancelled</code> flag checked before and during execution.	• Single vs Multi-Thread: Single timer thread delays subsequent tasks if one task hangs; Worker Pool decouples execution. • Fixed-Rate vs Fixed-Delay: Fixed-rate maintains absolute time schedule; Fixed-delay waits N ms after task completion.